Siemens ACM200 Axle Counter Count Discrepancy Troubleshooting

Diagnostic Procedure, Moisture Ingress Protocol, and Safety Constraints

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1. Count Discrepancy Alarm

A count discrepancy alarm is generated when the ACM200 evaluator detects that the number of axles counted entering a track section does not match the number counted leaving it. This condition means the system cannot confirm that the section is clear and will hold the section as occupied until the discrepancy is resolved by an authorised technician. Count discrepancies must be investigated and resolved before the section counter is reset.

WARNING: DO NOT reset the section counter to clear a count discrepancy without first physically confirming that no train or vehicle is present in the section. Resetting a counter while a train is occupying the section masks a genuine occupancy and can authorise a conflicting train movement - a potentially fatal error.

2. Initial Diagnostic Steps

Follow these steps in sequence. Stop and resolve if a cause is identified at any stage; do not proceed to later steps without clearing earlier ones.

- Step 1: Confirm with the control centre that no train or on-track machine is authorised in the section. Obtain a physical occupation check from the last signaller to handle train movements through the section.
- Step 2: Physically walk the section to confirm it is clear. This is mandatory before any counter reset is authorised.
- Step 3: Check sensor head alignment per MAN-AC-001 Section 3. Even minor misalignment can cause intermittent missed counts, especially at high speed.
- Step 4: Inspect the junction box for moisture ingress (see Section 3 below).
- Step 5: Review the evaluator event log for recurring discrepancy patterns. Single-event discrepancies may indicate a transient cause; recurring patterns at regular intervals suggest a timing or interference issue.

3. Moisture Ingress in Junction Box

Moisture in the sensor junction box is the most common non-alignment cause of count discrepancy alarms. Water ingress creates intermittent leakage paths that introduce noise into the sensor signal, causing spurious counts. Inspect as follows:

- Open the junction box lid. Inspect for visible water droplets, condensation, corrosion on terminals, or white mineral deposits (salt from evaporated water).
- If moisture is present: dry the interior with compressed air (clean, dry, oil-free). Replace the cable gland seals and junction box gasket using the standard seal kit (P/N ACM200-SEAL-KIT). Do not use silicone sealant as a substitute for proper gaskets.
- Inspect the cable entry glands. Loose or cracked glands are the most common ingress point. Replace glands if cracked or if the cable is not firmly gripped.
- After drying and resealing, monitor the system for 24 hours before closing the defect.
- If discrepancies persist after resealing, replace the sensor head unit (P/N ACM200-SH-ASSY) as an internal seal failure is likely.

Spare Part	Part Number	Notes
Junction box seal kit	ACM200-SEAL-KIT	Includes gasket, 4 cable gland seals
Sensor head assembly (contingency)	ACM200-SH-ASSY	Replace if moisture persists after resealing
M10 bolt set (stainless)	ACM200-M10-SS-4PK	Replace if corroded

NOTE: After any count discrepancy resolution, conduct a minimum 3-axle verification count before handing back *to the control centre. Document the cause, corrective action, and verification count result on the CMMS work order.*